

TE 458 — Coverage Planning: Tutorial 1 Practice Set (Variant B)

4 new numeric variants of Tutorial 1's problems (5G NR downlink/uplink, UMTS indoor, WCDMA macro-cell), same methods, new numbers

These are freshly parameterised variants of Tutorial 1's four problems — same methods and formulas (see the companion Formula Sheet), different input numbers — for additional self-test practice. Propagation-model coefficients are given directly in each question, exactly as they would be handed to you on an exam.

Problem 1 — 5G NR Downlink (variant)

In a 5G NR system operating at 2.1 GHz, the base station transmitter power is 44 dBm, with a feeder loss of 2.5 dB and an antenna gain of 17 dBi. For the UE receiver, the noise figure is 7 dB with a noise floor of -95 dBm, and the required SNR (E_b/N_0) for 16-QAM is 11 dB.

(Boltzmann's constant: $k = 1.38 \times 10^{-23}$ J/K, $T_0 = 290$ K. Additional margins: shadow-fading margin = 8 dB, interference margin = 4 dB. Propagation model: $MAPL = 132.4 + 37.2 \log_{10}(R)$.)

- Calculate the EIRP.
- Calculate the bandwidth.
- Compute the receiver sensitivity.
- Including the shadow-fading and interference margins above, find the Maximum Allowable Path Loss for the downlink.
- Using the given propagation model, determine the cell radius (R).

Problem 2 — 5G NR Uplink (variant)

A 5G NR uplink operates at $f = 2.1$ GHz. The user equipment transmits $P_{TX} = 100$ mW with antenna gain $G_{TX} = 0$ dBi and no feeder loss. The gNB receiver parameters are:

Parameter	Value
Receive antenna gain, G_{RX}	15 dBi
Receiver-side loss, L_{RX}	1.5 dB
Noise figure, NF	6.5 dB
Bandwidth, B	15 MHz

To support 64-QAM modulation, the required SNR is 15 dB. Additional losses and margins: body loss = 3 dB, building penetration loss = 8 dB, fade + interference margin = 10 dB. Assume $T_0 = 300$ K.

- Calculate the EIRP.
- Determine the free-space path loss at a reference distance of 100 m.
- Assuming $T_0 = 300$ K, determine: i. thermal noise spectral density; ii. thermal noise power in 15 MHz; iii. receiver noise floor; iv. receiver sensitivity.
- Compute the maximum allowable path loss (MAPL).
- Using the log-distance model with $n = 3.2$ and reference distance $d_0 = 100$ m, determine: i. the coverage radius; ii. the coverage area assuming a hexagonal cell.
- State and quantify the effect on coverage area if receiver sensitivity improves by 4 dB.

Problem 3 — UMTS System Design (variant)

As part of a UMTS system design, the following parameters are given to perform a link budget analysis for a 64-kbps indoor data service:

Parameter	Value
Mobile TX power (dBm)	21
Antenna gain (dBi)	0
Body loss (dB)	0
Thermal noise density (dBm/Hz)	-174
Receiver noise figure (dB)	6
Interference margin (dB)	3
Required E_b/N_0 (dB)	3
Base station antenna gain (dBi)	17
Base station feeder and connector losses (dB)	2.5
Fast fading margin (dB)	5
Log-normal fade margin (dB)	8
Building penetration loss (dB)	12
Soft handover gain (dB)	2.5
Chip rate, M_{cps}	3.84×10^6

- Perform link budget calculations for the 64 kbps indoor data service in the uplink direction for a laptop, considering 12 users.
- What is the maximum allowable path loss? (Propagation model: $\text{MAPL} = 137.4 + 35.2 \log_{10}(R)$.)
- What is the coverage radius of the base station?

Problem 4 — WCDMA Macro-Cell Analysis (variant)

You are planning the uplink link budget for a WCDMA macro-cell in a dense urban area supporting a 144 kbps video streaming service. Given:

Parameter	Value
Mobile transmit power	24 dBm
Mobile antenna gain	1 dBi
Body loss	3 dB
Base station receiver noise figure	5 dB
Chip rate	3.84 Mcps
Video service bit rate	144 kbps
Required E_b/N_0	5 dB
Thermal noise density	-174 dBm/Hz
Interference margin (60% cell loading)	6 dB
Slow fading margin (90% edge coverage probability)	8 dB

Soft handover gain	+2 dB
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- a. Compute the following step-by-step, showing all intermediate formulas: i. EIRP; ii. receiver noise power; iii. total noise-plus-interference power; iv. processing gain; v. effective receiver sensitivity; vi. Maximum Allowable Path Loss; vii. cell radius R using the propagation model $PL_{\max} = 135.0 + 36.0 \log_{10}(R)$.
- b. Perform a quantitative impact analysis on the hexagonal cell coverage area $A = (3\sqrt{3}/2)R^2$:
- If the service bit rate is increased to 288 kbps (same E_b/N_0), calculate the new radius and percentage reduction in coverage area.
 - If the interference margin rises to 9 dB due to higher network load, determine the new MAPL, radius, and area. Discuss the trade-off with system capacity.